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CSCE 580 Quiz 3

Q1: Search and Heuristics [2 + 2 + 2 + 4 = 10 points]

Instructions: Give your answers in bullet points.

- a. What is an admissible heuristics ? [2 points]
 - a. An admissible heuristic is one where the estimated cost is less than or equal to the actual cost. That is, it does not overestimate the cost.
- b. Suppose you are given “ $h = 0$ ” as the heuristics for a problem. Is it admissible ? [2 points]
 - a. Yes because for a heuristic to be admissible it must be less than or equal to the actual cost, and the actual cost cannot be less than 0.
- c. Suppose you are given “ $h = k$ ” as the heuristics for a problem, where k is any number e.g., $k=1$? Can you say it is admissible ? [2 points]
 - a. You cannot say it is admissible because if the true cost to goal $h^*(n)$ is greater than the heuristic $h(n)$ then it is not admissible. If $h^*(n) = 0$ and $h(n) = 1$ then it is not admissible. The only case where we know it is admissible is if $k = 0$.
- d. You are given 3 heuristics: h_1 , h_2 and h_3 of which you are sure that one is admissible. Will $h = \min(h_1, h_2, h_3)$ be admissible ? What can we say about $h = \max(h_1, h_2, h_3)$ as admissible ? [2 + 2 = 4 points]
 - a. $H = \min(h_1, h_2, h_3)$ will be admissible if we know that one is admissible. This is because an admissible heuristic must be less than or equal to the actual cost, so the minimum heuristic must be the admissible one.
 - b. $H = \max(h_1, h_2, h_3)$ is not necessarily admissible if we know that one is admissible. This is because an admissible heuristic must be less than or equal to the actual cost, so we know that the minimum of these heuristics are admissible but we cannot say the same for the maximum because they could be greater than the actual cost.

Q2.1 What state representation does it implement – express goal in it ? What search strategy does it implement ? [5 + 5 = 10 points]

- a. The state representation is: (left_missionaries, left_cannibals, right_missionaries, right_cannibals, boat_position)
- b. The goal state is: left_missionaries = 0, left_cannibals = 0, right_missionaries = initial_missionaries, right_cannibals = initial_cannibals, and boat_position = right.
- c. In other words, all entities have moved to the right and the boat is on the right.
- d. This code implements the breadth first search strategy.

Q3: Formulating a search problem [15 + 5 = 20 points]

a) Consider the cryptanalyst problem:

T W 0

+ T W 0

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Where each character stands for a unique number in the range (0-9). Formulate it as a CSP. What are the variables, their domains and constraints? [5 * 3 = 15 points]

- Variables: {T, W, O, F, U, R}
- Domains: {0,1,2,3,4,5,6,7,8,9}
- Constraints:
 - $O + O = R$
 - $T \neq 0$
 - $T \neq W, T \neq O, T \neq F, T \neq U, T \neq R,$
 - $W \neq O, W \neq F, W \neq U, W \neq R,$
 - $O \neq F, O \neq U, O \neq R,$
 - $F \neq U, F \neq R,$
 - $U \neq R$

b) Implement any of the non-search methods to simplify the formulation. Can we try to solve / simplify using node, arc or path consistency discussed in class? Show the closest you can come to a solution via pseudo-code? [5 points]

- Node consistency:

- function NC
 - for each V in nodes(G)
 - for each X in the domain D of V
 - if any unary constraint on V is inconsistent with X
 - Then delete X from D;